

Post-detection related developments, and SETI in Society news: 2023

(PC Members: please see full written report for details, links, etc.)

An update for the IAA SETI Permanent Committee, 4 Oct 2023, at IAC Baku

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General Status

- We have many members interested in post-detection / discovery & society, but we aren't currently formally organized into a PD working group within the Committee. Meanwhile, considerable research and discussion relevant to PD does continue (whirlwind review below).
- A small task group has started work specifically/narrowly on the groundwork for Protocols revisions (more in a moment).

Overarching issues to bear in mind

- Strong interest in SETI post-detection / 'discovery and society' topics continues, both within and beyond the Committee. n.b. post-detection-related research is essential, but persistently underfunded
- The Committee has much work ahead concerning Protocols updates: this is more complicated than last time...
- Larger contexts include: ongoing discussions about UAPS etc. (though generally outside this committee's remit); discussions in biosigs/technosigs community about standards of evidence, scientific consensus, and science communication; ongoing METI friction; mis/disinformation a serious and escalating concern for science and society

Overview of what will be briefly covered

- Papers on themes related to Post-detection in 2023 IACs
- Some relevant recent and upcoming activities of SETI PC members
- Protocols Revisions: Work in progress

1. Papers related to Post-detection: IAC 2023

<https://iafastro.directory/iac/browse/IAC-23/A4/>

- **Lori Walton:** "Billingham Cutting Edge Lecture" - The History of the IAA SETI Permanent Committee - 1990 to 1999
- **Kate Genevieve *et al*:** The SETI Post-Detection Hub: Preparing Humanity for Contact
- **Carol Oliver *et al*** (presented by Les Tennen): Revising the SETI Post-Detection Protocols for the 2020s and Beyond: A Report on Work in Progress
- **Kate Genevieve *et al*:** The ocean calls - SETI, lunar astronomy and scenarios at the 21-cm hydrogen line
- **Daniela de Paulis:** Composition, decoding and interpretation of the simulated extraterrestrial message in the interdisciplinary project "A Sign in Space".
- **Gabriela Radulescu:** Regulating Outer Space: Contact with Extraterrestrial Intelligence in the International Astronautical Community in the late 1950s

1. Papers related to Post-detection: IAC 2023, cont'd

<https://iafastro.directory/iac/browse/IAC-23/A4/>

- **Paolo Musso *et al***: Astrobiology and SETI in Peru: recent and future advances
- **Paolo Musso *et al***: Insights for SETI from linguistics studies in the Peruvian Amazon
- **Debarshi Mukherjee**: Societal Implications of Extraterrestrial Intelligence: Public Reaction, Risk Communication, and Possible Impacts on Society
- **Ugur Guven**: How to Distinguish a Potential Spaceship in Search for Extraterrestrial Intelligence?
- **Suresh Bhattarai *et al***: Are we alone? People's perception on an ExoWorld, its planet and possible life on it

2. Some relevant recent and upcoming activities of SETI Permanent Committee members

- N.b. This compilation is not comprehensive. Please do share your updates/information with me.
- **Continued public outreach by SETI Institute and Breakthrough Listen, continued public engagement by other SETI projects**
- **Technical work on technosignature candidate followup**
- **SETI Post-Detection Hub (John Elliott will say more):** The Post-Detection Hub, which started as a UKSRN initiative, launched in Nov 2022. Jointly hosted by the St Andrews Centre for Exoplanet Science and the Centre for Global Law and Governance , with additional affiliates.
- **PSETI:** seminar series, and PSETI Symposium 2023 talks and breakout session “The Alien in the Machine: Post-Detection in the Era of AI”
- **Daniela De Paulis, A Sign in Space (Daniela will say more)**
- Institute for Ascertaining Scientific Consensus, Durham
- **Upcoming:** IAU/Kavli Symposium, “(Toward) Discovery of Life Beyond Earth and its Impact” (Durham/virtual 2024); NASA Astrobiology Program workshop “Communicating Discoveries in the Search for Life in the Universe” (virtual, Feb 2024)

3. The Protocols: work in progress

- A small task group (Carol Oliver, Kathryn Denning, Michael Garrett, Les Tennen) produced a paper for IAC Baku re: the preliminaries for a revision of the Protocols. Committee members: please read it.
- **Key points:**
 - Preliminary consultations with some SETI researchers gave us some idea of what we need to work on, and we sketch that in the paper. There will be more opportunities for SETI community input.
 - We will need your collegiality, patience, and expertise. Thanks to those who have contributed already.
 - If you'd like to learn more about the Protocols' history... contact me at kdenning@yorku.ca and I'll share resources.
 - The initial formulation of the Protocols (1989) took years, and the 2010 revision took years. And our situation is more complicated now.
 - We are aiming for: streamlined timeless Principles; revised Preamble; auxiliary Codes/Best Practices which are more easily updated; groundwork for discussion on bigger issues which can't be handled effectively within a Principles revision (METI)
- More to come. Thanks!